Web Application Development

Lab Assignment – 7

Online Food Ordering Application Using Vue JS Routing and State Management

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Objective

To develop an Online Food Ordering application using Vue 3. The application uses Vue Router with nested routing for the Order section, and Pinia for managing the food cart state, including item quantities and totals.

Project Structure

```
food-ordering/
├── package.json
├── vite.config.js
├── index.html
├── README.md
└── src/
    ├── main.js
    ├── App.vue
    ├── router/
    │   └── index.js
    ├── stores/
    │   └── cart.js
    └── views/
        ├── Home.vue
        ├── Order.vue
        ├── Menu.vue
        ├── Cart.vue
        └── About.vue
```

Routing / Component Flow

```
App.vue
  ├── Home            →  /
  ├── Order           →  /order
  │     ├── Food Menu  →  /order/menu
  │     └── Cart       →  /order/cart
  └── About           →  /about
```

Order.vue acts as the parent route and contains its own <router-view>. Menu.vue and Cart.vue are nested child routes of /order, so they are displayed inside the Order page itself.

Implementation

main.js

Entry point of the application. It creates the Vue app, installs Vue Router and Pinia, and mounts the app to #app.

```
import { createApp } from 'vue'
import { createPinia } from 'pinia'
import App from './App.vue'
import router from './router'
```

```
createApp(App)
  .use(router)
  .use(createPinia())
  .mount('#app')
```

App.vue

Root component. It contains the main navigation (Home, Order, About) using router-link, and a router-view to display the matched page.

```
<template>
  <div class="app">
    <nav class="navbar">
      <router-link to="/">Home</router-link>
      <router-link to="/order">Order</router-link>
      <router-link to="/about">About</router-link>
    </nav>

    <router-view />
  </div>
</template>

<script>
export default {
  name: 'App'
}
</script>

<style>
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
}

.app {
  max-width: 700px;
  margin: 0 auto;
  padding: 15px;
}

.navbar {
  display: flex;
  gap: 15px;
  padding-bottom: 15px;
  margin-bottom: 15px;
```

```
  border-bottom: 1px solid #ccc;
}

.navbar a {
  text-decoration: none;
  color: #2c3e50;
  font-weight: bold;
}

.navbar a.router-link-exact-active {
  color: #e67e22;
}

button {
  cursor: pointer;
}
</style>
```

index.js

Configures Vue Router. Defines routes for /, /order, /about, and nests /order/menu and /order/cart as child routes of /order.

```
import { createRouter, createWebHistory } from 'vue-router'
import Home from '../views/Home.vue'
import Order from '../views/Order.vue'
import Menu from '../views/Menu.vue'
import Cart from '../views/Cart.vue'
import About from '../views/About.vue'

const routes = [
  { path: '/', component: Home },
  {
    path: '/order',
    component: Order,
    children: [
      { path: 'menu', component: Menu },
      { path: 'cart', component: Cart }
    ]
  },
  { path: '/about', component: About }
]

const router = createRouter({
  history: createWebHistory(),
  routes
})

export default router
```

stores/cart.js

Pinia store that maintains the cart state. Provides actions addToCart, increaseQuantity, decreaseQuantity, removeFromCart and clearCart, and getters for totalItems and totalBill.

```
import { defineStore } from 'pinia'

export const useCartStore = defineStore('cart', {
  state: () => ({
    items: []
  }),

  getters: {
    totalItems(state) {
      return state.items.reduce((sum, item) => sum + item.quantity, 0)
    },
    totalBill(state) {
      return state.items.reduce((sum, item) => sum + item.price * item.quantity, 0)
    }
  },

  actions: {
    addToCart(food) {
      const existing = this.items.find(item => item.id === food.id)
      if (existing) {
        existing.quantity++
      } else {
        this.items.push({ ...food, quantity: 1 })
      }
    },
    increaseQuantity(id) {
      const item = this.items.find(item => item.id === id)
      if (item) {
        item.quantity++
      }
    },
    decreaseQuantity(id) {
      const item = this.items.find(item => item.id === id)
      if (item) {
        item.quantity--
        if (item.quantity <= 0) {
          this.removeFromCart(id)
        }
      }
    },
    removeFromCart(id) {
      this.items = this.items.filter(item => item.id !== id)
    },
    clearCart() {
      this.items = []
    }
  }
```

```
})
```

Home.vue

Simple welcome page displayed at the / route.

```
<template>
  <div>
    <h1>Welcome to Food Order</h1>
    <p>Order your favorite food easily.</p>
  </div>
</template>

<script>
export default {
  name: 'Home'
}
</script>
```

Order.vue

Parent route for the order section. Contains links to Food Menu and Cart, and a nested router-view that displays Menu.vue or Cart.vue.

```
<template>
  <div>
    <h2>Order</h2>
    <nav class="order-nav">
      <router-link to="/order/menu">Food Menu</router-link>
      <router-link to="/order/cart">Cart</router-link>
    </nav>

    <router-view />
  </div>
</template>

<script>
export default {
  name: 'Order'
}
</script>

<style scoped>
.order-nav {
  display: flex;
  gap: 15px;
  margin-bottom: 15px;
}
```

```
.order-nav a {
  text-decoration: none;
  color: #2980b9;
}
</style>
```

Menu.vue

Displays the four food items (Pizza, Burger, Pasta, Coke) using v-for. Each item has an Add to Cart button that calls the Pinia addToCart action.

```
<template>
  <div>
    <h3>Food Menu</h3>
    <div class="food-item" v-for="food in foods" :key="food.id">
      <span>{{ food.emoji }} {{ food.name }}</span>
      <span>₹{{ food.price }}</span>
      <button @click="cart.addToCart(food)">Add to Cart</button>
    </div>
  </div>
</template>

<script>
import { useCartStore } from '../stores/cart'

export default {
  name: 'Menu',
  data() {
    return {
      foods: [
        { id: 1, name: 'Pizza', price: 250, emoji: '🍕' },
        { id: 2, name: 'Burger', price: 150, emoji: '🍔' },
        { id: 3, name: 'Pasta', price: 200, emoji: '🍝' },
        { id: 4, name: 'Coke', price: 80, emoji: '🥤' }
      ]
    }
  },
  computed: {
    cart() {
      return useCartStore()
    }
  }
}
</script>

<style scoped>
.food-item {
  display: flex;
  align-items: center;
  gap: 15px;
```

```
  padding: 10px 0;
  border-bottom: 1px solid #eee;
}

.food-item span:first-child {
  flex: 1;
}

button {
  padding: 5px 10px;
  background-color: #27ae60;
  color: white;
  border: none;
  border-radius: 4px;
}
</style>
```

Cart.vue

Displays the cart items from the Pinia store with quantity, remove and clear controls, shows the total bill, and provides a Place Order button that clears the cart and shows a success message.

```
<template>
  <div>
    <h3>🛒 Cart ({{ cart.totalItems }} items)</h3>

    <p v-if="cart.items.length === 0">Your cart is empty.</p>

    <div v-else>
      <div class="cart-item" v-for="item in cart.items" :key="item.id">
        <span>{{ item.emoji }} {{ item.name }}</span>

        <button @click="cart.decreaseQuantity(item.id)">-</button>

        <span>{{ item.quantity }}</span>

        <button @click="cart.increaseQuantity(item.id)">+</button>

        <span>₹{{ item.price * item.quantity }}</span>

        <button @click="cart.removeFromCart(item.id)">Remove</button>
      </div>

      <hr />

      <p><strong>Total: ₹{{ cart.totalBill }}</strong></p>

      <button @click="cart.clearCart()">Clear Cart</button>

      <button @click="placeOrder">Place Order</button>
```

```
      <p v-if="orderMessage" class="success">
        {{ orderMessage }}
      </p>
    </div>
  </div>
</template>

<script>
import { useCartStore } from '../stores/cart'

export default {
  name: 'Cart',

  data() {
    return {
      orderMessage: ''
    }
  },

  computed: {
    cart() {
      return useCartStore()
    }
  },

  methods: {
    placeOrder() {
      this.orderMessage = 'Order placed successfully!'
      this.cart.clearCart()
    }
  }
}
</script>

<style scoped>
.cart-item {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 8px 0;
  border-bottom: 1px solid #eee;
}

.cart-item span:first-child {
  flex: 1;
}

button {
  padding: 4px 8px;
  border: 1px solid #ccc;
  background-color: #f5f5f5;
```

```
  border-radius: 4px;
}

.success {
  margin-top: 15px;
  font-weight: bold;
}
</style>
```

About.vue

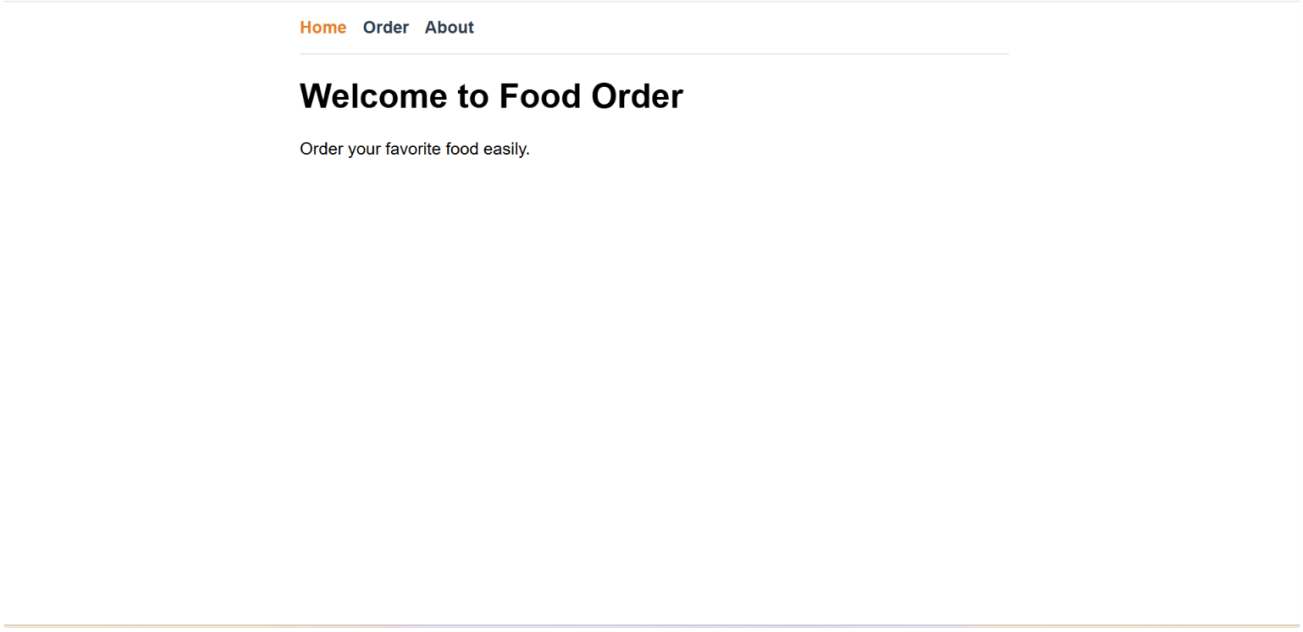
Displays a short description of the application at the /about route.

```
<template>
  <div>
    <h2>About</h2>
    <p>
      This application is a simple online food ordering system
      developed using Vue.js, Vue Router and Pinia.
    </p>
  </div>
</template>

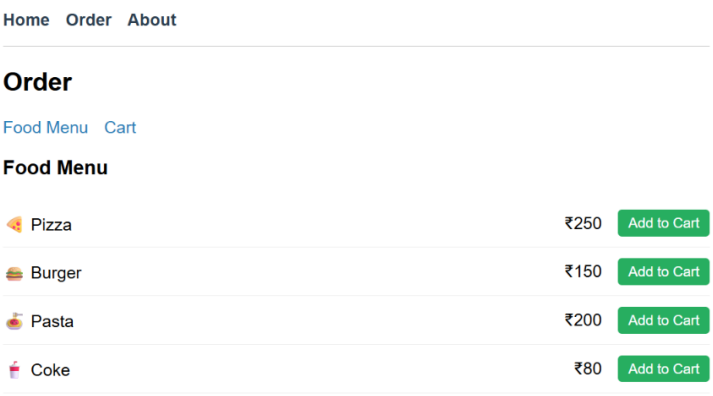
<script>
export default {
  name: 'About'
}
</script>
```

Screenshots:

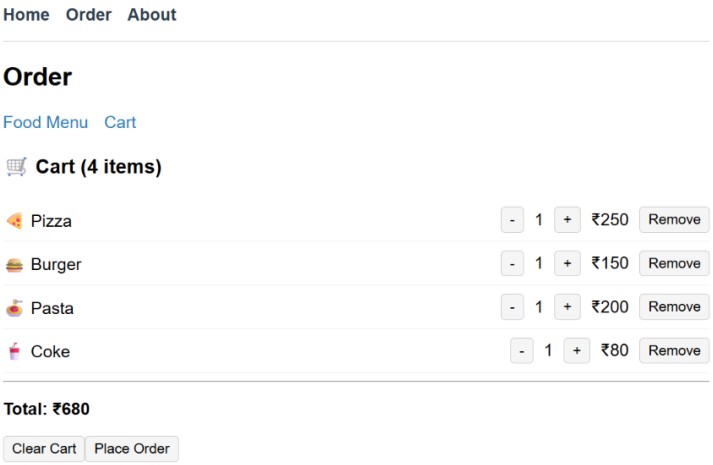
Home Page:



Menu Page:



Cart Page:



Learning Outcomes

- I learned how to use Vue Router to navigate between different pages of a Vue application.
- I learned how to implement nested routing to display the Food Menu and Cart pages inside the Order page.
- I learned how to use Pinia for state management to handle cart operations and calculate totals.